SUNIL BELDE

(945)-244-4371 sunil.belde102@gmail.com ◇ LinkedIn GitHub Portfolio Medium

EDUCATION

University of Illinois Chicago , *Chicago, IL*
M.S in Computer Science *Aug 2022 – May 2024*

VNR VJIET , *Hyderabad, India*
B.Tech in Electronics and Communication Engineering *Aug 2015 – Jun 2019*

TECHNICAL SKILLS

Programming Languages: Python, Java, Pyspark, SQL, Bash
Machine Learning & GenAI: Transformers, TensorFlow, PyTorch, LLMs (Llama, Mistral), Prompt Engineering, RAG
Data Engineering & Pipelines: LlamaIndex, LangChain, SimSearch, MLflow, DVC, Jinja2, NetworkX
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, Glue, CloudWatch), GCP, Docker, Kubernetes, Git, CI/CD
Backend & Web Development: Spring Boot, REST APIs, MySQL, Angular 8
Databases & Storage: PostgreSQL, MongoDB, Elasticsearch, Vector Databases
Tools & Platforms: Jupyter Notebook, VS Code, Tableau, Kibana, Prodigy

EXPERIENCE

Capital One, ML Engineer – GenAI/LLM Automation, *Chicago, IL* Jul 2024 – Present

- Developed a human-in-the-loop Conversion Mapping Engine using the **ReAct agentic AI pattern** to propose, critique, and refine source-to-target mappings for the Discover File System (DFS) migration, grounded in requirements documents, cross-organization discussions, and historical conversion knowledge.
- Integrated **enterprise MCP tools and plugins**, including Exchange, Glean, and Ask-Data, enabling the agent to discover registered data assets, retrieve enterprise knowledge, and query data context to inform mapping decisions.
- Evaluated the assistant on cross-domain benchmarks, achieving **75% source-field recall**; assessed mapping accuracy across structural correctness, source-field selection, and transformation conditions.
- Architected a **GenAI ETL platform** that translates natural-language mapping instructions into **PySpark code** deployable as **AWS Glue jobs**, reducing engineering effort by over **70%** and delivery time from approximately **5 months to 4 weeks**.
- Combined **Llama 3.3-70B**, **LangChain**, and **SimSearch**-based RAG with deterministic **Python/NetworkX DAG orchestration**, automated code validation, and feedback-driven retry loops for ETL generation and assembly.
- Developed an **enterprise data-quality (EDQ) workflow** using **Claude Skills and automation scripts** to generate conversion validation rules, produce **pass/fail reports**, and support **human review** for the Discover File System migration.

University of Illinois Chicago, Graduate Research Assistant, *Chicago, IL* Jan 2023 – May 2024

- Built end-to-end **NLP pipelines** using **HuggingFace Transformers**, **SpaCy**, and **scikit-learn** to analyze political trends from **10M+ tweets**; insights contributed to published academic research.
- Implemented scalable preprocessing modules including **scraping**, **text normalization**, **sentiment analysis**, **topic modeling** (LDA), and visualization with **Pandas, Matplotlib, Seaborn**.

Dockskiff – A Jaggaer Company, Machine Learning Engineer, *Hyderabad, India* Jul 2020 – Jul 2022

- Built **document intelligence pipelines** combining **NER**, **OCR**, and NLP to extract contract metadata using **PyTorch**, **Tesseract**, and regex-based postprocessing.
- Increased classification throughput by **30%** through document segmentation using **Faster-RCNN**.
- Enhanced OCR quality with custom **image preprocessing** techniques (denoising, deskewing) to improve text accuracy.
- Developed a **Clause Classification Engine** using **Legal-BERT** and **FAISS-based semantic search** to identify and compare clauses for compliance and risk analysis.

Acheron Software Consultancy Pvt Ltd, Associate Software Developer, *Hyderabad, India* Jan 2019 – Apr 2020

- Transitioned from a **DevOps Intern** to full-time engineer; contributed to app development, deployment automation, and infrastructure monitoring.
- Built enterprise-grade applications using **Spring Boot**, **MySQL**, and **Angular 8**, delivering production-ready backend APIs and responsive UIs.
- Set up a centralized **observability stack** using the **ELK suite** (Elasticsearch, Logstash, Kibana) on **GCP**, improving logging and diagnostics.
- Migrated legacy systems to **Kubernetes** using **Helm**, reducing deployment overhead and cutting infra costs by **15%**.

CERTIFICATIONS

AWS Solutions Architect – Associate Issued by Amazon Web Services
Applied AI – Machine Learning Certification Apr 2020 – Jun 2021
Building Chatbots with Watson and Python (IBM) Issued by IBM